

SZÉCHENYI ISTVÁN EGYETEM

Rally Fuel Level Sensor Selection

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Brief:

Select and justify a fuel level measurement system for a rally car, from the perspective of the system engineer responsible for delivering it to the powertrain team.

Fundamentals of fuel level measurement, and road car appliances

In a road vehicle, monitoring the fuel level is essential for safe and reliable operation. It allows the driver to plan refuelling in advance, avoid unexpected breakdowns, and ensure uninterrupted travel.

In a race car, it is also essential to be aware of the current fuel level, as the driver, or co-driver may need to adapt their driving/navigation style accordingly to ensure that enough fuel remains in the car until the end of the race. Managing throttle application, lift-and-coast techniques, and overall pace can be influenced by real-time fuel information. Without accurate data, there is a significant risk of running out of fuel before the finish.

In my case, the task was to measure the fuel level. In rally racing, engineers typically calculate the required fuel quantity in advance to make sure the driver can complete the planned stage distance. However, having a reliable fuel level measurement system provides an additional layer of safety and verification, ensuring that the pre-calculated amount is sufficient under real conditions, and later the data gathered from the fuel level measurements can be used to analyse engine performance, and improve efficiency.

At their core, **fuel level sensors are devices designed to measure the amount of fuel in a tank and relay this information to a display system**. A fuel level sensor consists of two main components: **the sensing system itself** (also known as the sender) and the **indicator** (also commonly referred to as the gauge). The market offers various fuel level sensor types, each with distinct operating principles, advantages, and limitations.

 Float arm Float moves an arm that changes resistance. + Simple, cheap, reliable - Wear, slosh-sensitive Road cars	 Resistive Fuel level changes electrical resistance. + Cost-effective, moderate accuracy - Slosh-sensitive, electrical wear Passenger vehicles	 Selected Capacitive Measures capacitance change along the probe. + No moving parts, high accuracy - Fuel-type dependent, higher cost High performance	 Ultrasonic Uses sound reflection time to measure level. + Non-contact, accurate - Foam/turbulence sensitive, complex Commercial fleets
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1. Table: Fuel Level Measurement Technologies

Technical limitations

To select the most suitable sensor type for rally racing, we must evaluate which technology performs best under rally-specific conditions and carefully analyse the challenges encountered during competition.

In terms of fuel level measurement, **fuel slosh** is very important. Fuel slosh is the uncontrolled movement of fuel inside a tank caused by acceleration, braking, cornering, or vertical impacts. This effect can lead to **unstable fuel readings, temporary fuel starvation, or even inconsistent pressure at the injectors**. One alternative solution to this is applying swirl pots.

Swirl Pot is a small auxiliary fuel reservoir used in motorsport fuel systems to prevent fuel starvation under high G-forces. But with a swirl pot, you **lose space, add weight to the car, and you must implement a new part** to your car, which is not ideal, and might be unnecessary.

As mentioned in the figure above, **Float arm sensors, and Resistive fuel level sensors are very sensitive to fuel slosh, so these two technologies are not optimal in a rally car**, although they offer a cheaper alternative, which can be considered if you would like to build a customer rally car, or a hobby car. However, for a car that would like to compete for championships, and race podiums these small sloshes can cost dearly.

Another key constraint in rally fuel system design is **the geometry and packaging of the fuel tank**. Benchmarking shows that most rally teams install the fuel cell **inside the vehicle, typically aligned with the rear axle**. This location supports optimized weight distribution and helps maintain a stable centre of gravity throughout the stage.

Rectangular or regular-shaped tanks are generally preferred. From a system engineering perspective, **simpler geometries improve sensor calibration accuracy, simplify volume-to-level mapping, reduce production complexity, and ease integration within the chassis**. Typical fuel cell capacities in WRC applications are 60 l.

In WRC rally regulations, **fuel quantity is controlled and verified in kilograms**. However, the **fuel level sensor measures volume in liters**. Since fuel mass depends on its density, and density changes with temperature, **weather conditions become a critical factor**. As temperature increases, fuel expands and its density decreases, meaning the same mass occupies a larger volume. Therefore, when converting from liters to kilograms, temperature must be considered to ensure **accurate fuel calculation** and compliance with regulations.

Structural robustness is a primary requirement. Rally stages expose the vehicle to rocks, jumps, compressions, and potential underbody impacts. The fuel cell must therefore withstand significant mechanical loads and deformation without rupture, minimizing the risk of leakage during accidents. For this reason, the tank is positioned inside the vehicle rather than underneath it.

Modern rally fuel cells are also filled with **safety foam**. This foam reduces fuel slosh, stabilizes fuel pickup under high longitudinal and lateral accelerations, and suppresses vapor expansion. In the event of an impact, it mitigates explosion risk by limiting rapid

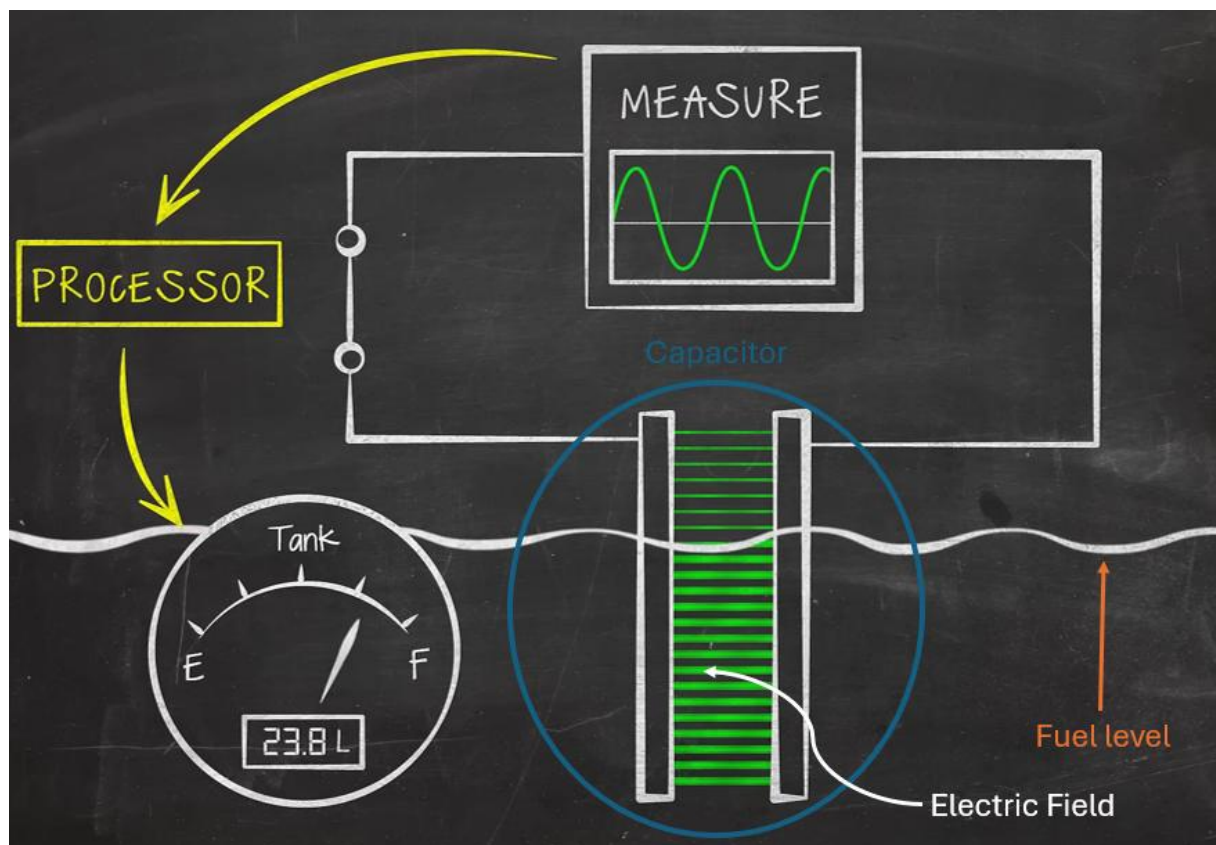
fuel movement and reducing the available volume for vapor ignition, thereby enhancing both operational reliability and safety under extreme conditions.

Fuel type and fuel characteristics are also very important. At the latest Acropolis rally, the temperatures were high (more than 40°C), and this caused vaporization inside the fuel tank, therefore a new fuel supplier had to step in, so the participants could race. The new synthetic fuel offers various challenges, and the engineers need more time to fully understand the characterization of the fuel. In case of **capacitive fuel sensors**, the fuel specifications must be accurate, so the sensor can produce precise data.

Due to their durability and precision, capacitive sensors are the optimal choice for fuel level monitoring under rally conditions.

Capacitive fuel sensors

Definition: Rather than physical contact with the fuel, these sensors **detect changes in electrical capacitance caused by varying fuel levels. Without moving parts, capacitive fuel sensors offer excellent durability and precision**, making them ideal for harsh environments.



1. Figure: Capacitive Fuel Level Sensor Functional Illustration

Performance Comparison of Selected Fuel Level Sensors

Based on diligent observation and comparative benchmarking, three capacitive fuel level sensors suitable for high-performance rally applications were identified. The subsequent decision matrix provides an analytical comparison of their technical specifications and operational tradeoffs. The first possible sensor is from [ATL, EL-AD-535](#), Skoda rally cars use them for fuel level measurements. The second is from [Gill, LevelLite](#) sensors, they are suppliers for Toyota Gazoo Racing, and the third is the [Reventec LS345](#).

Criterion · importance	ATL EL-AD-535	Gill LevelLite	Reventec LS345
Accuracy 5			
Operating temp. 4			
Sample rate 4			
Probe length 4			
Weight 3			
Mounting options 4			
Price 3			
Weighted score	47	62	61

2. Table: Weighted Decision Matrix

I considered accuracy as the most important factor. If we would like to monitor the situation of the fuel, which is crucial in motorsport, then we need proper data to work with. In this section the Reventec were the best option. According to the brochure it can work with only 1 % error rate, at 20°C, and for extra money they can reduce that to even 0,5 %. Meanwhile the Gill sensor can only guarantee the 1 % until 50°C. The operating temperature is close between the Gill and reventech, both can work in the same operating range (-40°C to 125°C). Meanwhile the ATL can only operate between -20 to 85°C. The sample rate is 100 Hz for the Gill and Reventech, meanwhile for the ATL it is only 80 Hz. In terms of mechanical attributes, the ATL can offer the best versions, the only part where they miss out is with the mounting options, they can't offer different variations. Where the ATL really excels at is the price category, the ATL EL-AD-535 costs only 255 €. The Gill, LevelLite cost approximately 650€, at the same time the basic Reventec LS 345 is around 850€.

Ultimately the ATL sensor can offer a cheap alternative, where accuracy is not that important and for customer racing it can be an ideal option. The Gill and Reventec sensors are very close to each other in terms of performance, and overall package. The only small difference where the **Gill LevelLite** excels at is the [documentation and support provided](#). This was not listed in the decision matrix, as this topic is a bit more subjective, and you can't really grade it, but it is also very important. It can help to integrate the system faster and therefore have a more reliable system.

Sensor placement

It is advised that the sensor should be placed on **top of the fuel tank**. This configuration ensures **accessibility** for inspection, emergency intervention, and recalibration without removing the tank from the vehicle. In addition, a top-mounted installation significantly **improves mechanical protection**, as the sensor is less exposed to debris impact, underbody strikes, and structural deformation during heavy landings or accidents.

From a structural perspective, the upper surface of the fuel cell is subjected to lower impact loads compared to the bottom and side walls. This **reduces mechanical stress** on the probe and sealing interface, decreasing the likelihood of leakage or sensor failure.

Top installation also **simplifies sealing** and improves reliability of the electrical connection, as cable routing can be kept above the fuel level, minimizing the risk of fuel ingress into connectors.

Finally, vertical top mounting allows the probe to follow the full usable fuel height, **ensuring maximum measurement range and more predictable calibration** characteristics.

Sampling Frequency and Measurement Range

The Gill LevelLite sensor is capable of a 100 Hz sampling frequency. However, such a **high data rate is not necessary for fuel level monitoring**, as the actual fuel mass in the tank does not change at such a rapid rate. While **transient slosh effects** may introduce **short-term fluctuations**, the overall fuel consumption is a relatively slow process. Therefore, a lower effective logging frequency between 1-10 Hz combined with appropriate **signal filtering** is sufficient, **reducing unnecessary data storage while maintaining accurate and reliable measurement**.

The selected fuel tank has a total capacity of **60 litres and a simple rectangular geometry**. Therefore, the measurement range of the sensor must be calibrated to accurately cover the full 0–60 L volume. Due to **the regular shape of the tank, the relationship between fuel height and volume is linear**, which simplifies calibration and improves measurement accuracy across the entire range. The Gill LevelLite sensor provide an [installation manual](#). To optimize the measurement range, the calibration process should be the following: **Full tank measurement → Empty tank measurement → Measurement with 1l → Measurement with 2l... and so on until it is measured to 60l fuel**